UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of
Artificial Intelligence

TOPIC 1:
INTRODUCTION TO DATA AND DATA WRANGLING

DATA WRANGLING AND VISUALIZATION (SAIA2143)

Prepared by:
Ts. Dr. Norziha Megat Mohd Zainuddin
norziha.kl@utm.my

TOPIC LEARNING OUTCOMES

At the end of this class, students should be able to:

1. Explain the Importance of Data Quality.
2. Identify the Dangers of Messy Data.
3. Define the concept and significance of data wrangling.
4. Describe the the key steps in Data Wrangling Process.
5. Recognize Tools for Data Wrangling.
6. Identify the Importance of Clean Data in Analysis.
7. Identify Different Types of Messy Data.
8. Analyze the Real-World Impact of Unclean Data.
9. Discover the Strategies to Prevent Data Quality Issues.

Introduction to Important of Data Quality

- Data is the bedrock for analysis and innovation.
- Data-driven in organizations are more likely to acquire and retain customers.
- Data helps identify trends and predict outcomes.

The Dangers Of Messy Data

1. **Inaccurate Insights**: Poor data quality leads to flawed decisions.
2. **Costly Errors:** Poor data quality costs businesses billions annually.
3. **Real-World Impact:** Healthcare errors and financial losses are caused by bad data.





CLASS ACTIVITY:
The Impact of Command Challenges of Messy Data

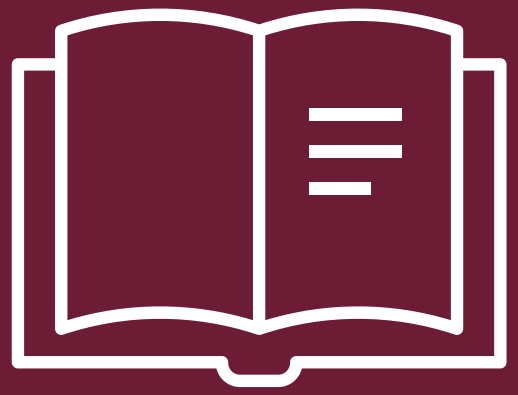
Instructions:

- Think of a time when incorrect or messy data caused a problem.
- It could be in school, work, shopping, banking, or daily life.
- Share:
  - What was the **messy data issue**?
  - What **problems** did it cause?
  - How was it **fixed (if at all)?**



 **Discussion:** Be ready to share with the class!

More reading



Article

Data Wrangling in Database Systems: Purging of Dirty Data

Otmane Azeroual

German Center for Higher Education Research and Science Studies (DZHW), Schützenstraße 6a, Berlin 10117, Germany; azeroual@dzhw.eu

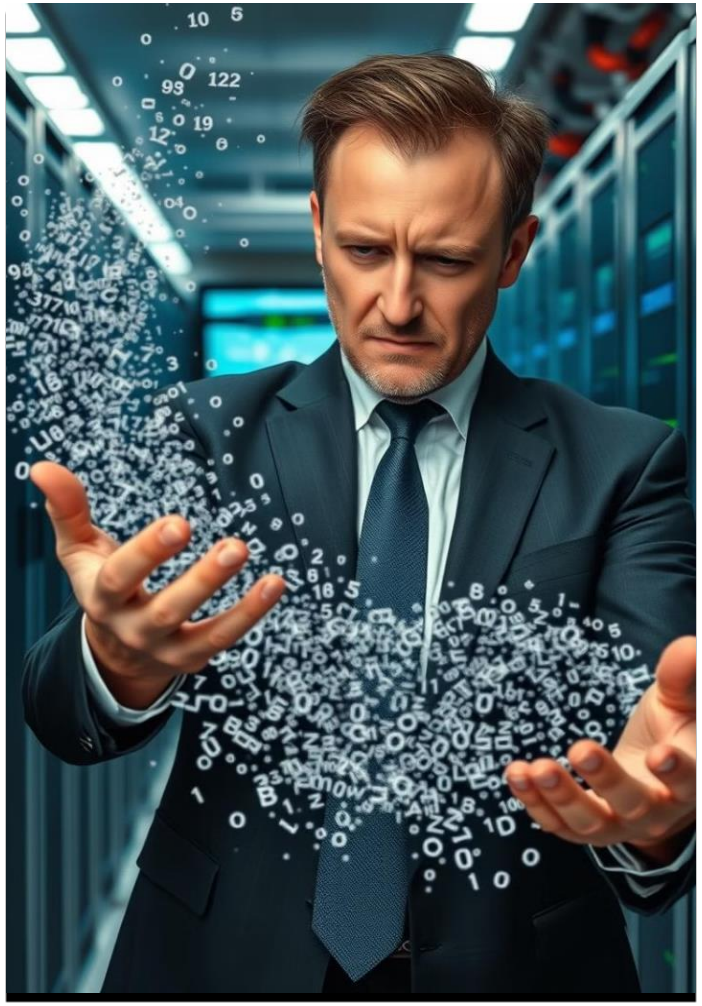
Received: 30 March 2020; Accepted: 2 June 2020; Published: 5 June 2020

Abstract: Researchers need to be able to integrate ever-increasing amounts of data into their institutional databases, regardless of the source, format, or size of the data. It is then necessary to use the increasing diversity of data to derive greater value from data for their organization. The processing of electronic data plays a central role in modern society. Data constitute a fundamental part of operational processes in companies and scientific organizations. In addition, they form the basis for decisions. Bad data quality can negatively affect decisions and have a negative impact on results. The quality of the data is crucial. This includes the new theme of data wrangling, sometimes referred to as data munging or data crunching, to find the dirty data and to transform and clean them. The aim of data wrangling is to prepare a lot of raw data in their original state so that they can be used for further analysis steps. Only then can knowledge be obtained that may bring added value. This paper shows how the data wrangling process works and how it can be used in database systems to clean up data from heterogeneous data sources during their acquisition and integration.

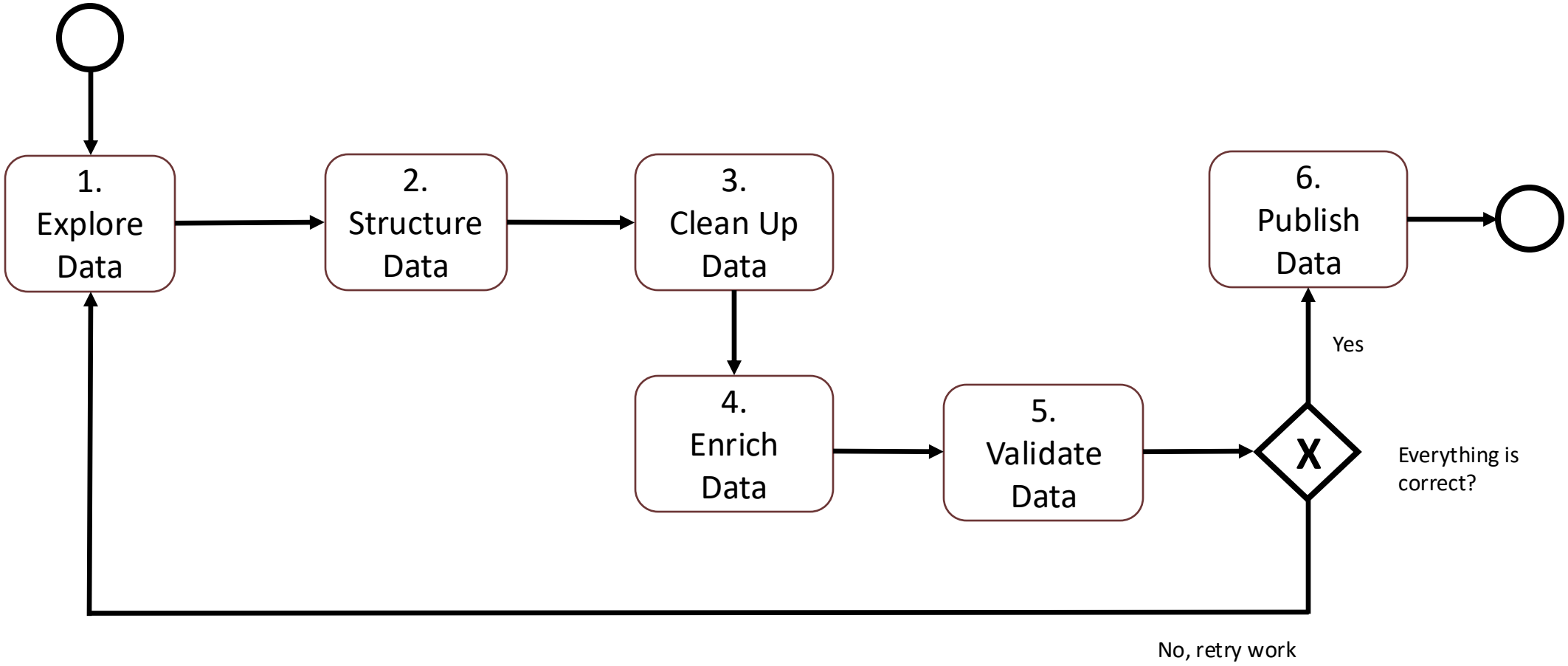
Keywords: information systems; data management systems; heterogeneous data; data integration; dirty data identification; data quality; data curation; data management; data wrangling; data munging; data crunching

Data Wrangling: Defining

- Data wrangling transforms messy data into a usable format.
- Protects against corrupt data.
- Important when merging databases.
- It accounts for most of the time spent on data science projects.



The Data Wrangling Process



Azeroual, O. (2020). Data wrangling in database systems: purging of dirty data. *Data*, *5*(2), 50.

Key Steps in Data Wrangling

1: Exploring

- Understand data origin and structure.
- Identify patterns, missing values, and inconsistencies.
- Essential for determining cleaning and transformation needs.

2: Structuring

- Convert unstructured data into a structured format.
- Remove unnecessary columns, split values, and standardize formats.
- Ensures consistency for accurate analysis.

3: Cleaning

- Detect and correct errors (e.g., duplicates, outliers, missing values).
- Standardize data formats for better accuracy.
- Enhances data quality and usability.

Key Steps in Data Wrangling…

4: Enriching

- Add new data from internal or external sources.
- Derive new values from existing data (e.g., date conversion, probability measures).
- Enhances dataset value for deeper insights.

5: Validating

- Check data quality and consistency across records.
- Ensure values follow defined rules and constraints.
- Prevents errors from affecting analysis outcomes.

6: Publishing

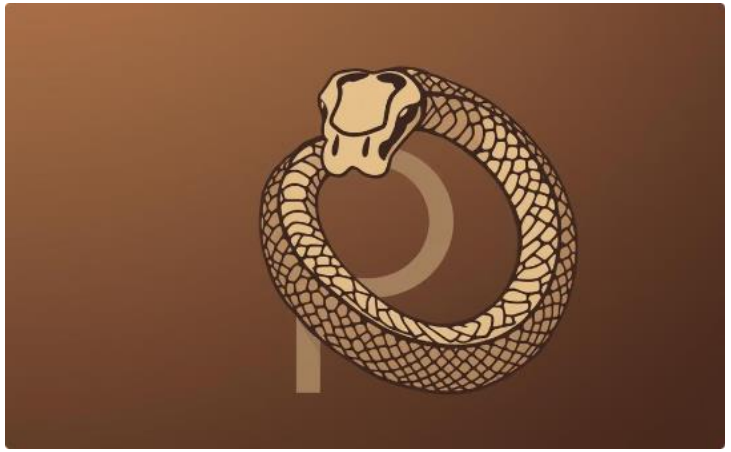
- Prepare and transfer cleaned data for analysis.
- Load data into relevant software or databases.
- Document the transformation process for future use.

Tools For Data Wrangling



Spreadsheet Software

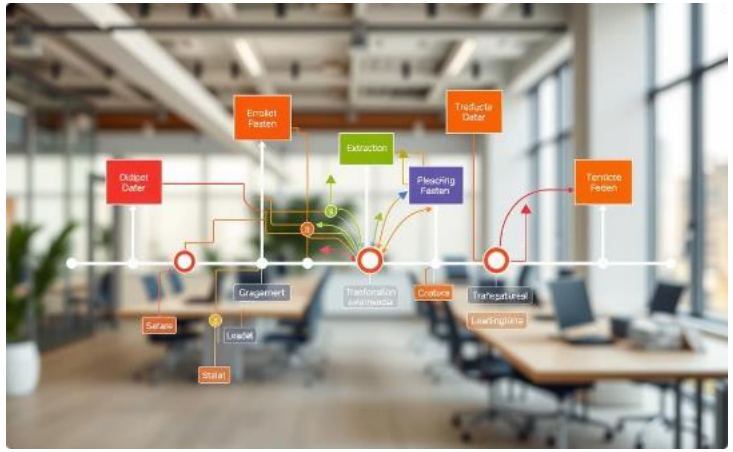
Basic cleaning tasks.



Programming Languages

Powerful libraries for data manipulation.

Example: Python and R



ETL Tools

Large-scale data integration.

Example: OpenRefine, Trifacta and Alteryx

TRIFACTA

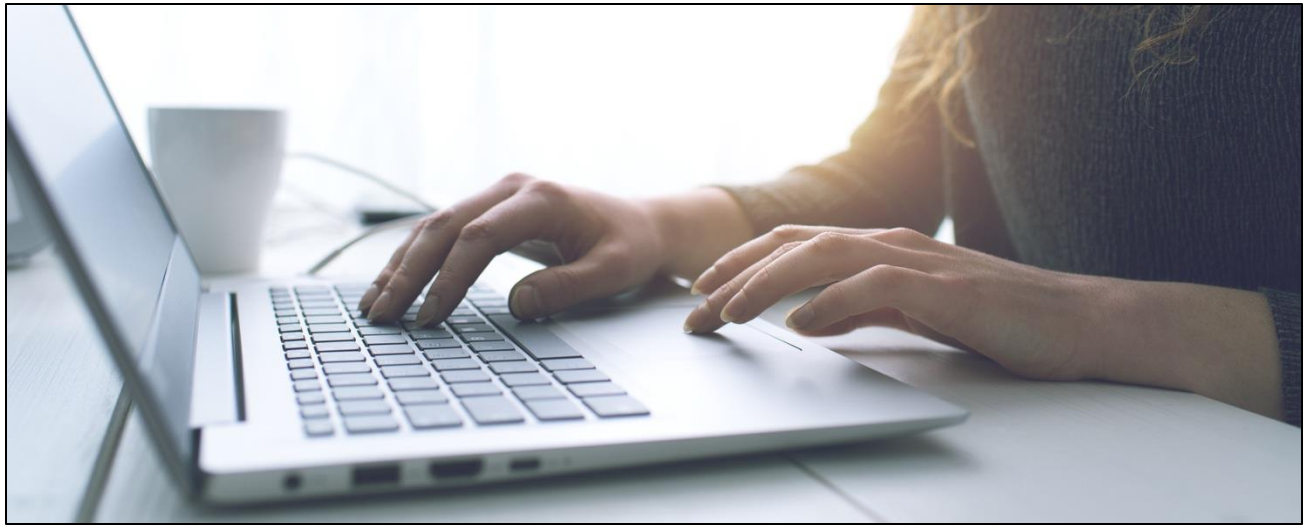
Importance Of Clean Data In Analysis

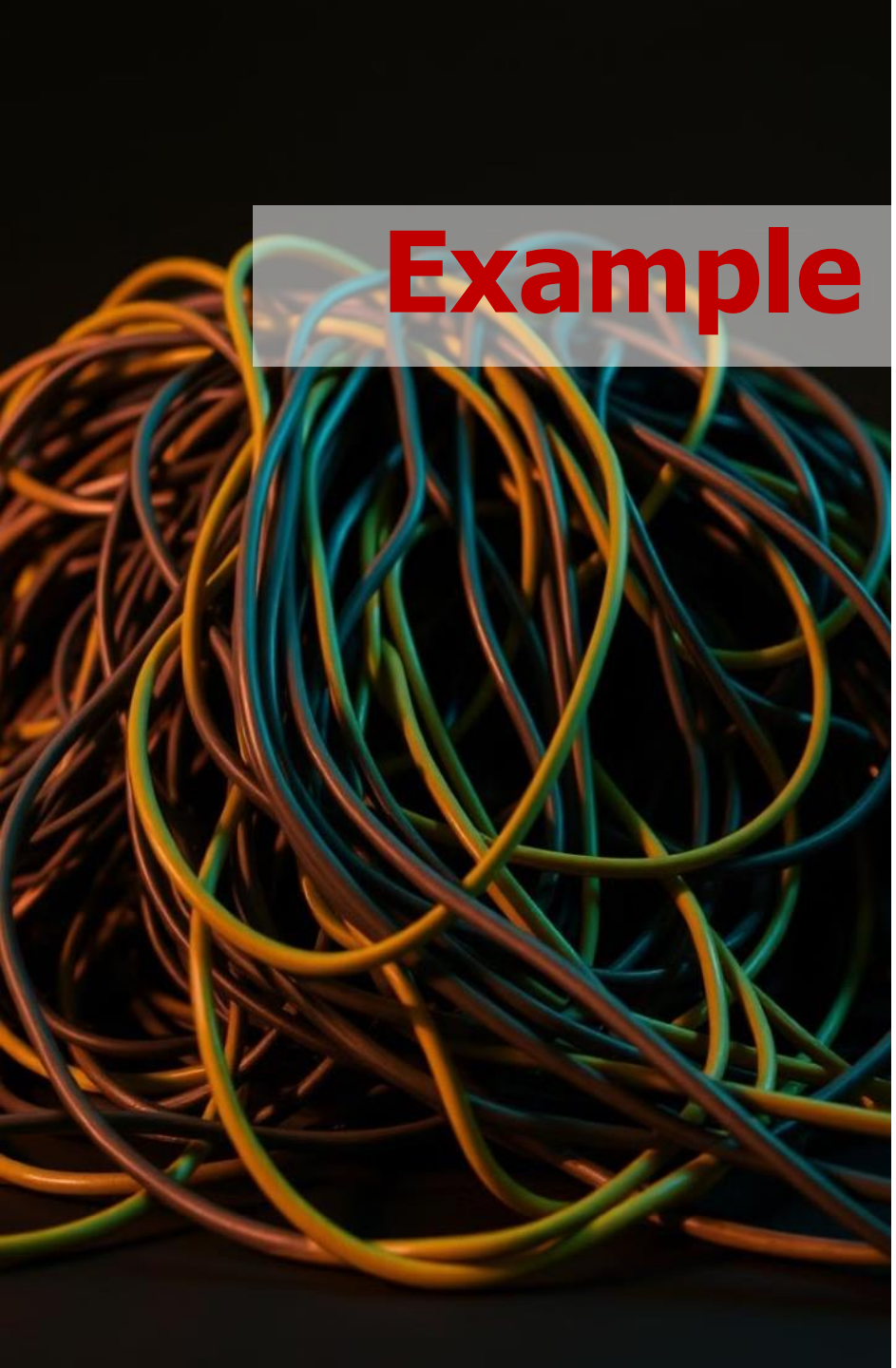
1. **Accuracy:** Ensures correct analysis and decision-making.
2. **Consistency:** Eliminates redundancy and contradictions.
3. **Efficiency:** Reduces time spent on fixing errors later.
4. **Compliance:** Meets regulatory and ethical standards.



Types Of Messy Data

- Not maintained attributes
- Abuse of attributes for additional information
- Incorrect data caused by incorrect input, including, e.g., wrong reading, etc.
- Typing error
- Inaccurate data
- Missing data
- Redundant and inconsistent data
- Various incorrect formats
- Duplicate records
- Outdated information





Example : Types Of Messy Data

Incomplete - Missing values or fields in CRM data.
(e.g., missing customer phone numbers).

Inconsistent - Conflicting formats.
(e.g., date formats varying between MM/DD/YYYY and DD/MM/YYYY).

Inaccurate - Errors, typos, or outdated info.
(e.g., outliers in sales data).

Duplicate - Redundant entries in the dataset.
(e.g., duplicate customer entries).

Real-World Example Impact of Unclean Data on Decision-Making

- **Business Failures:** Unreliable data leads to incorrect financial forecasting, affecting company revenue and operations.

- **Healthcare Risks:** Incorrect or outdated patient records may result in medical misdiagnoses and improper treatments.

- **Operational Inefficiencies:** Errors in inventory management due to bad data can cause supply chain disruptions.

Preventing the Negative Impact of Unclean Data

- **Data Validation & Cleaning:** Regular audits and automated cleaning processes can improve data reliability.
- **Training & Awareness:** Educating employees on data accuracy reduces human errors.
- **Standardization & Governance:** Establishing consistent data entry standards enhances data integrity.
- **Technology & Tools:** Using AI and machine learning to detect anomalies in datasets.

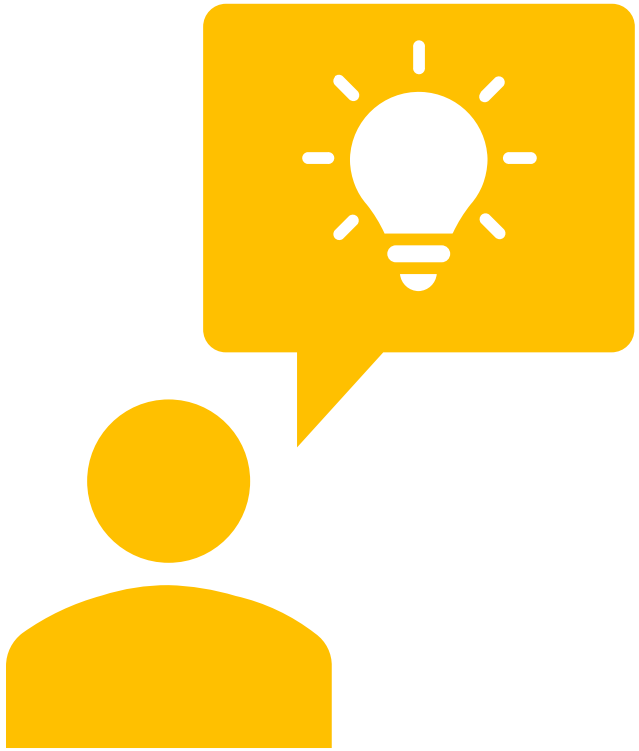


CLASS ACTIVITY

Based on the supply chain management issues caused by unclean data please discuss in group the issue on the:

1. **Inventory Mismatches:** How can inaccurate stock levels impact business operations, and what strategies can be used to prevent overstocking or stockouts?
2. **Delayed Shipments:** What are the possible consequences of incorrect order data on delivery timelines, and how can companies improve data accuracy in logistics?
3. **Supplier Miscommunication:** How can inconsistent supplier records affect procurement decisions, and what measures can businesses take to maintain accurate supplier data?
4. **Demand Forecasting Errors:** How does unreliable sales data lead to poor demand predictions, and what data quality practices can improve forecasting accuracy?
5. **Cost Overruns:** In what ways can data errors increase operational costs in supply chain management, and how can businesses reduce financial risks associated with poor data quality?

KNOWLEDGE CHECK



THANK YOU



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